

ZINDER, Niger

WMO: 610900

Lat: 13.779N

Lon: 8.984E

Elev: 1516

StdP: 13.91

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	57.2	59.1	17.2	13.8	82.0	17.2	13.9	82.2	20.7	71.8	18.9	71.8	5.8	40	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	22.6	107.3	66.6	105.8	67.0	104.3	67.2	78.0	91.4	77.2	90.3	76.5	89.5	7.5	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.1	139.3	82.1	74.1	134.8	82.0	73.5	131.8	81.8	42.6	91.6	41.8	90.3	41.2	89.6	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.5	16.7	15.3	DB	52.1	110.2	2.2	1.3	50.6	111.1	49.3	111.8	48.1	112.5	46.5	113.4
			WB	39.8	81.6	0.8	2.0	39.2	83.0	38.8	84.2	38.3	85.3	37.7	86.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	84.2	72.4	79.5	86.1	92.0	93.3	90.4	85.5	82.6	85.5	87.4	81.0	74.2
	DBStd	7.42	4.99	5.65	4.38	3.33	3.05	3.82	3.81	3.22	3.26	2.43	3.74	4.15
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1	1	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12467	694	826	1120	1261	1342	1211	1100	1010	1064	1160	929	750
	CDD65	6993	230	406	655	811	877	761	635	545	614	695	479	285
	CDH74	95711	2333	4979	9418	13147	14273	11348	8101	6010	7739	9754	5801	2808
	CDH80	56837	978	2719	5831	9012	9886	7225	4268	2693	4110	5829	3125	1161
Wind	WSAvg	7.6	8.5	8.0	8.3	7.4	7.6	8.8	9.1	7.1	5.7	6.0	7.0	8.0
Precipitation	PrecAvg	16.2	0.0	0.0	0.0	0.0	0.7	1.6	4.9	6.4	2.3	0.4	0.0	0.0
	PrecMax	25.8	0.0	0.2	0.7	0.7	3.5	3.9	11.5	13.8	6.8	6.3	0.2	0.0
	PrecMin	8.6	0.0	0.0	0.0	0.0	0.0	0.0	0.7	0.0	0.3	0.0	0.0	0.0
	PrecStd	4.4	0.0	0.0	0.1	0.1	0.7	1.1	2.2	2.8	1.5	0.9	0.0	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.8	102.6	105.8	109.2	109.1	106.2	102.0	96.8	102.0	103.2	100.2	95.4
		MCWB	60.4	62.9	63.5	65.0	67.6	71.4	73.4	75.5	68.8	65.8	63.1	60.7
	2%	DB	93.1	100.1	103.9	107.4	107.5	104.4	98.9	94.8	99.4	101.8	98.2	92.5
		MCWB	58.5	61.6	62.7	65.6	67.2	71.8	74.3	75.9	71.8	66.1	62.1	59.2
	5%	DB	89.5	97.0	102.1	105.9	106.0	102.6	96.8	93.0	97.2	100.3	95.6	89.7
		MCWB	57.0	59.9	62.0	65.5	67.7	72.4	74.5	75.7	73.6	66.3	61.1	57.5
	10%	DB	86.0	93.9	100.1	104.2	104.4	100.6	94.7	91.1	95.2	98.6	93.5	86.8
		MCWB	55.7	58.9	61.0	65.6	68.2	72.8	74.6	75.5	74.5	66.5	60.5	56.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.9	64.8	69.6	75.1	77.2	78.5	78.9	78.7	79.1	77.3	68.3	62.4
		MCDB	93.2	98.2	95.4	98.6	96.5	96.5	92.6	90.6	91.5	90.8	92.8	90.0
	2%	WB	59.5	62.9	66.1	73.3	75.6	77.0	77.2	77.5	77.8	75.7	64.5	60.2
		MCDB	90.1	96.2	97.2	96.5	95.6	94.6	90.2	89.1	90.3	90.1	93.5	90.1
	5%	WB	57.6	60.9	63.5	71.9	74.6	76.0	76.4	76.9	77.0	74.5	62.7	58.6
		MCDB	87.8	94.4	98.0	94.7	94.8	93.5	89.2	88.1	89.5	89.0	92.8	87.2
	10%	WB	55.9	59.3	61.6	70.3	73.7	75.3	75.8	76.3	76.3	73.1	61.1	56.9
		MCDB	85.4	92.2	97.2	94.4	93.8	92.5	88.6	87.0	88.6	87.5	91.0	84.9
Mean Daily Temperature Range	5% DB	MDBR	24.5	25.6	25.3	24.6	22.6	20.3	17.8	16.1	19.4	23.2	25.1	24.4
		MCDBR	29.1	28.6	28.1	25.7	24.5	22.1	20.2	18.4	22.0	25.2	27.2	27.7
		MCWBR	11.8	10.7	10.5	10.4	10.5	7.0	5.2	5.4	6.8	8.7	9.4	10.9
	5% WB	MCDBR	28.5	27.4	26.3	23.1	21.5	20.3	17.9	16.9	19.4	21.5	26.5	26.9
		MCWBR	12.5	11.4	11.1	11.1	8.9	6.8	5.8	5.9	6.7	8.4	10.3	11.3
Clear-Sky Solar Irradiance	taub	0.557	0.617	0.670	0.722	0.709	0.700	0.649	0.613	0.618	0.653	0.556	0.546	
	taud	1.710	1.567	1.478	1.425	1.465	1.528	1.676	1.750	1.721	1.584	1.684	1.721	
	Ebn at Noon	235	227	220	209	207	207	219	230	229	216	233	233	
	Edh at Noon	73	88	98	103	97	90	78	74	76	85	75	71	
All-Sky Solar Radiation	RadAvg	1795	1942	2060	2114	2097	2002	1917	1885	2003	1943	1838	1735	
	RadStd	59	51	87	71	73	78	56	62	48	59	51	64	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page